

1 PROBLEM & GOAL

Complex technical research requires an agent to autonomously decompose queries, explore live web endpoints, recover from unexpected execution faults, filter noise, and compile structured synthesis reports. This project implements a production-grade **Autonomous Web Research Agent** that takes a high-level research goal, executes dynamic multi-step search and retrieval, self-corrects on failures, deduplicates heterogeneous evidence, and outputs an evidence-backed markdown report without human intervention.

2 ARCHITECTURE & DESIGN DECISIONS

The system enforces a strict separation of concerns: **LLMs reason and plan, while deterministic application code orchestrates execution, enforces security boundaries, and tracks state.**

- Reasoning Backbone (Gemini 3.5 Flash-Lite):** Selected for sub-second structured generation, high context capacity, and cost-effective multi-turn reasoning.
- Closed Planning Schema:** The Gemini Developer API rejects OpenAPI schemas containing open-ended `additionalProperties`. To ensure robust native structured output, we decoupled the LLM planning contract (`LLMPlan`, `PlannedTask`, `ToolInput` with explicit typed arguments) from internal runtime entities (`Plan`, `Task`).
- LangGraph Cyclic Graph:** Built on `ResearchAgentState` using typed channels. Bounded cyclic loops support dynamic replanning, progressive evidence accumulation, and stateful trace logging.
- Tool Registry Whitelist:** Enforces exact schema validation on tool invocations (`web_search` via Tavily, `page_fetcher` with HTTP pooling). Arbitrary execution is strictly prohibited.
- Evidence Pipeline:** Extracted text is segmented into paragraphs, ranked by lexical query keyword density (threshold ≥ 0.30), and deduplicated via exact SHA-256 and token Jaccard similarity (≥ 0.75).

4 SECURITY & ROBUSTNESS

- SSRF Firewall:** The page fetcher intercepts URL requests at the DNS resolution level, blocking RFC 1918 private subnets (`10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`), loopback (`127.0.0.0/8`, `:::1`), link-local (`169.254.0.0/16`), and cloud metadata endpoints.
- Payload Caps & Budgets:** Hard limit of 2MB per HTTP response, 50,000-character content truncation, and a ceiling of 10 maximum execution steps prevent memory exhaustion and runaway billing.
- Zero-Leak Credential Safety:** API secrets are injected strictly via environment variables, excluded from git tracking via `.gitignore`, and sanitized from execution traces and logs.

7. ARCHITECTURAL PHILOSOPHY & SUMMARY

Reliable autonomy is achieved not by granting an LLM unconstrained execution privileges, but by establishing strict deterministic boundaries: closed schema validation, rule-based state machine routing, autonomous failure recovery paths, and defense-in-depth security guardrails.

3 AUTONOMOUS RECOVERY & FAILURE HANDLING

Tool execution is treated as inherently non-deterministic. A rule-based `Evaluator` inspects step outcomes and transitions the state across a deterministic decision matrix:

Decision	Trigger Condition	Action Taken
CONTINUE	Step succeeded with valid evidence	Advances plan task index
RETRY	Transient error (timeout, 5xx, parse)	Retries task up to 2 times
REPLAN	Persistent error (404), zero evidence	LLM generates revised subtasks
COMPLETE	All tasks done or evidence satisfied	Routes to report synthesis
FAIL	Execution limit (≥ 10 steps) tripped	Halts execution to prevent loops

- Empirical Recovery Validations:**
- Timeout Retry:* Simulated network timeouts triggered automatic retry cycles, recovering with complete evidence on attempt 2.
 - 404 Dynamic Replanning:* Live encounters with dead URLs triggered immediate replan transitions, discarding dead links and generating fresh search queries without human intervention.

5 TESTING & VALIDATION STRATEGY

- 66/66 Automated Tests Passing:** Complete test suite runs offline in ~3.0s using mocked LLM and search protocols with fault-injection fixtures.
- End-to-End Live Validation:** Validated against live Gemini 3.5 Flash-Lite and Tavily Search APIs on complex agentic AI research topics.
- Regression Guarding:** Verifies planning schemas, SSRF blocking, retry bounds, deduplication thresholds, and report formatting.

6 KNOWN LIMITATIONS & FUTURE IMPROVEMENTS

- Lexical vs. Dense Similarity:** Current deduplication and relevance use lexical Jaccard/frequency; dense vector embeddings would improve semantic nuance.
- Single-Agent Topology:** Upgrading to a multi-agent supervisor pattern would enable concurrent specialized crawling and fact-checking subagents.
- Interface Modality:** Terminal CLI-first; exposing an SSE streaming API with a React web dashboard would enhance real-time observability.